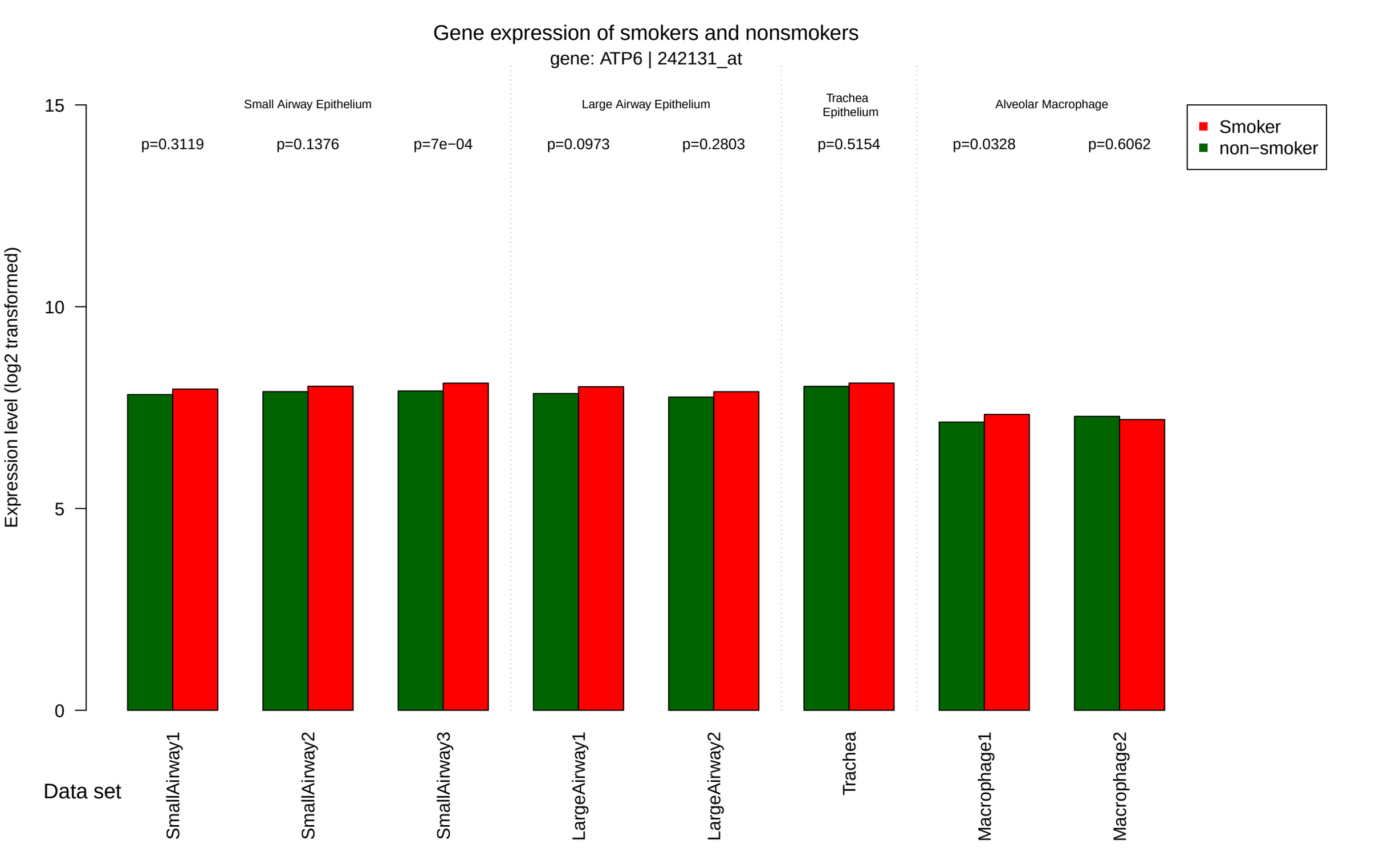
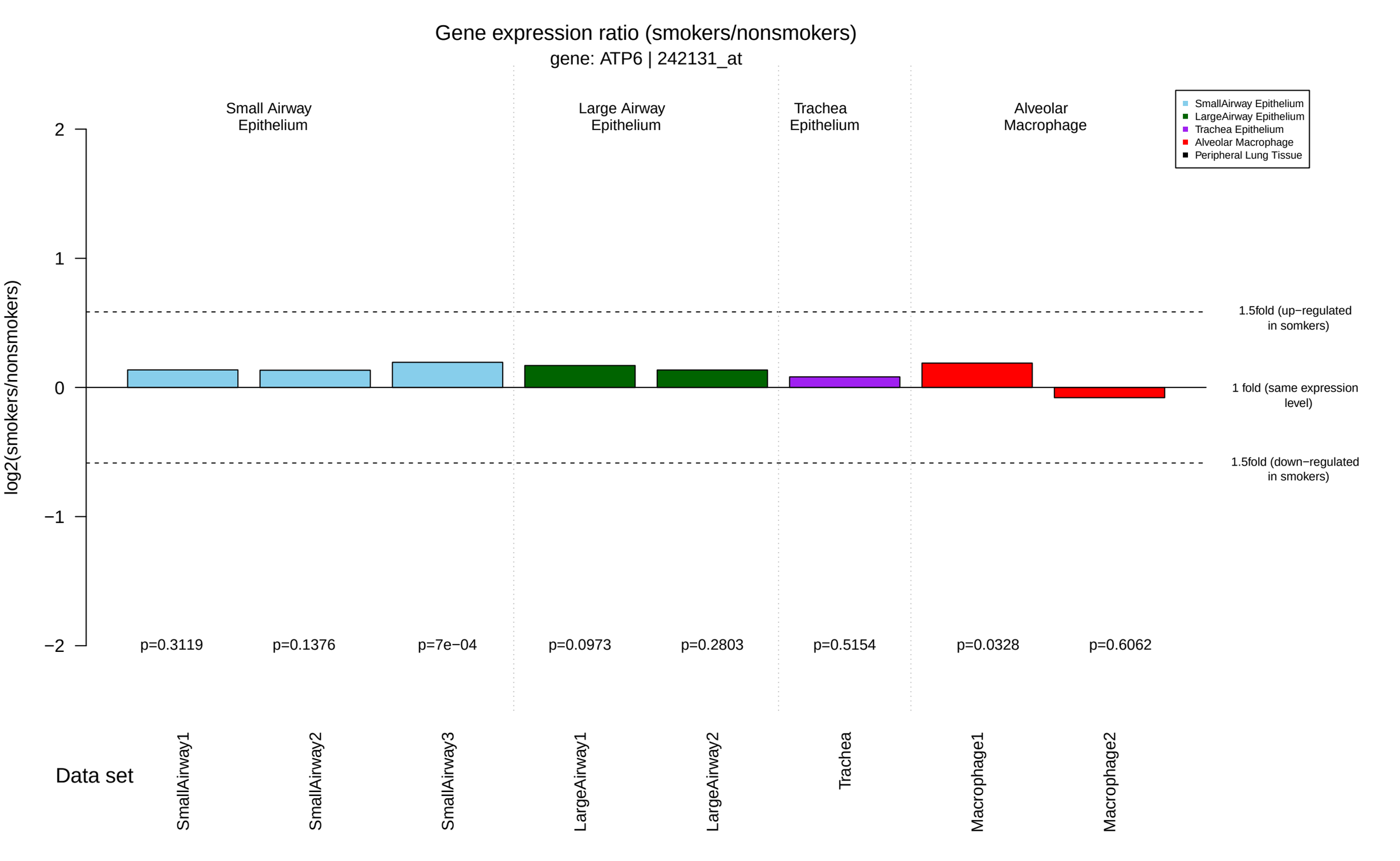
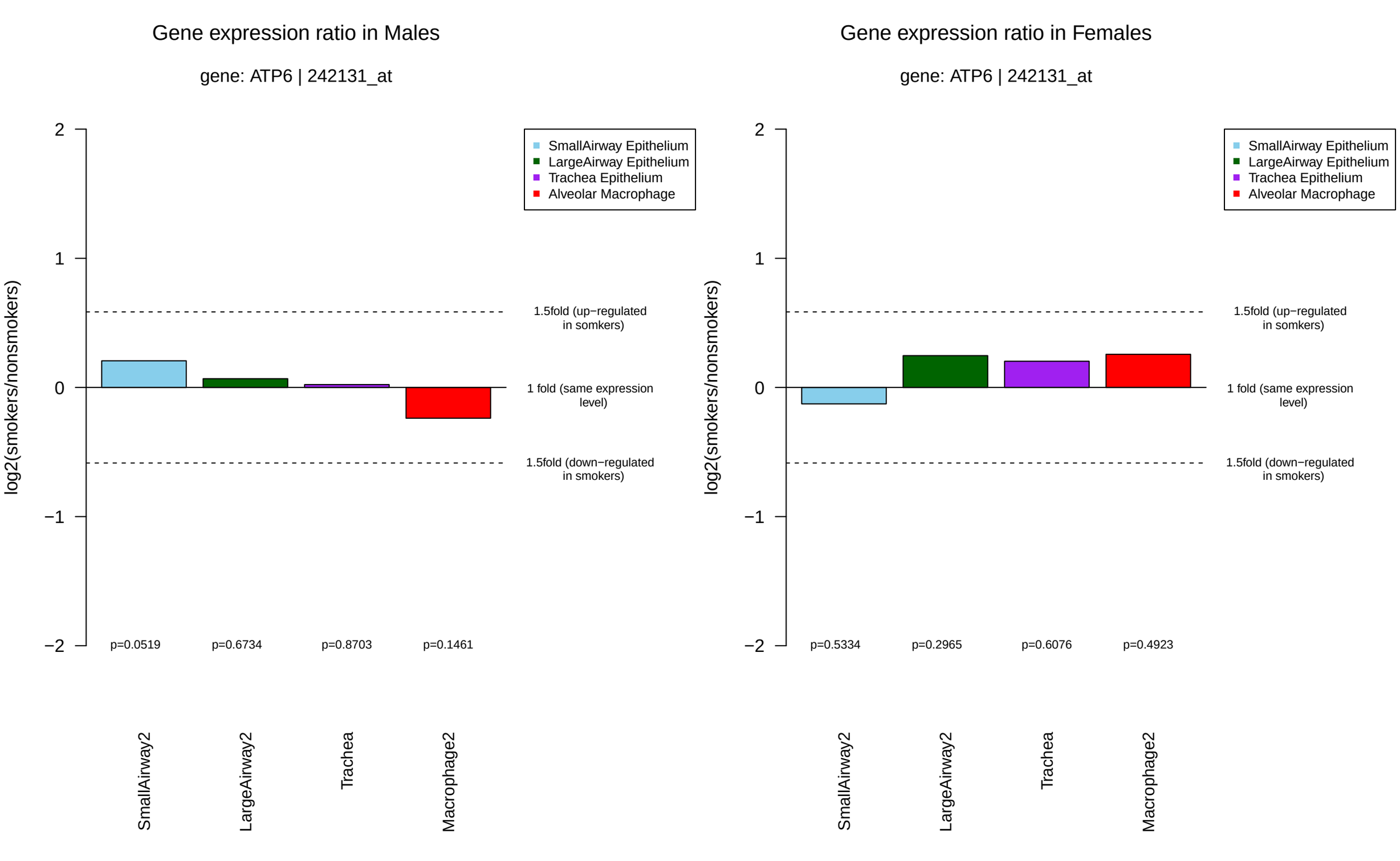
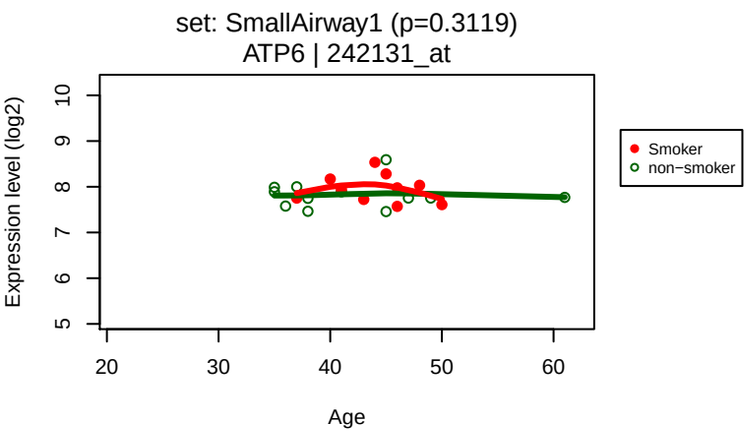
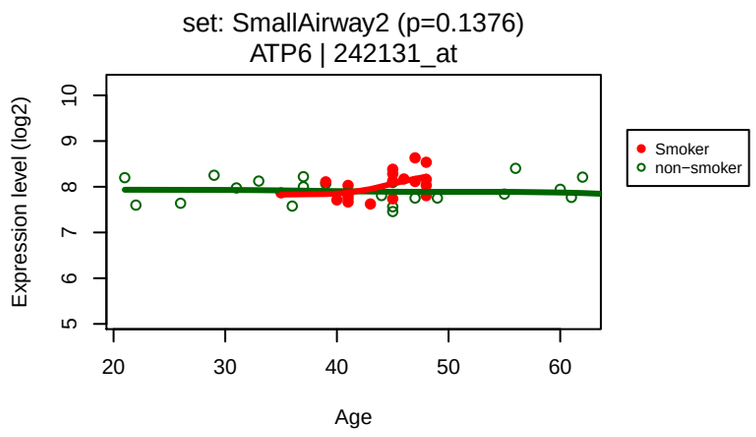
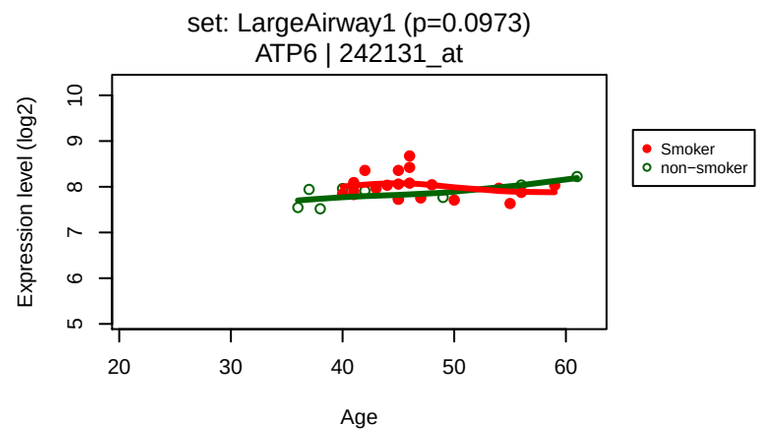

Gene expression of smokers and nonsmokers
gene: ATP6 | 242131_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3119
p=0.1376
p=7e−04
p=0.0973
p=0.2803
p=0.5154
p=0.0328
p=0.6062
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

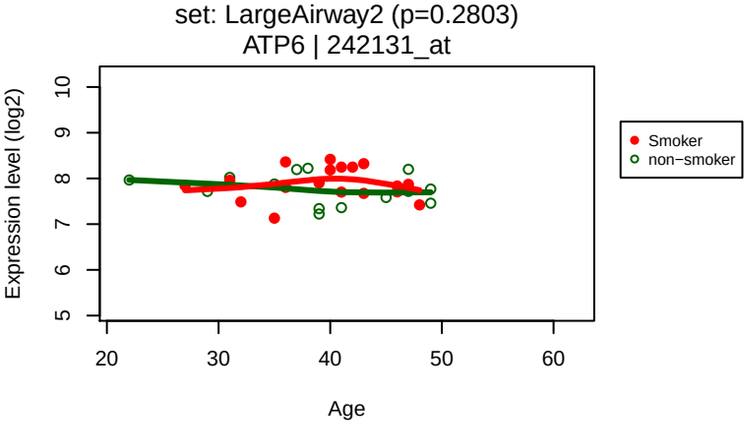

Gene expression ratio (smokers/nonsmokers)
gene: ATP6 | 242131_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3119
p=0.1376
p=7e−04
p=0.0973
p=0.2803
p=0.5154
p=0.0328
p=0.6062
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

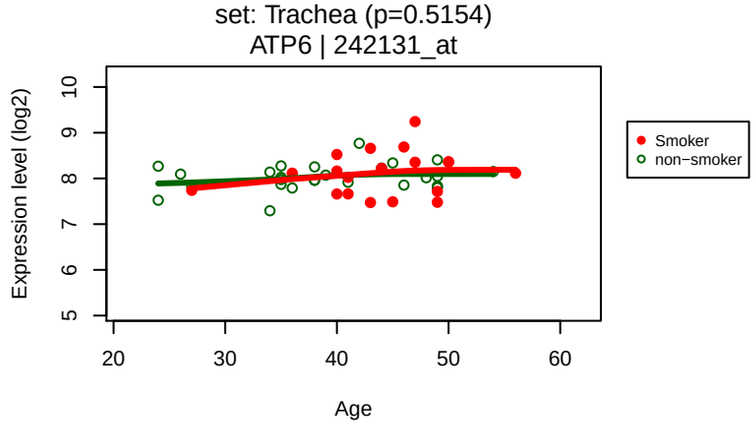

Gene expression ratio in Males
gene: ATP6 | 242131_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0519
p=0.6734
p=0.8703
p=0.1461
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP6 | 242131_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5334
p=0.2965
p=0.6076
p=0.4923
SmallAirway2
LargeAirway2
Trachea
Macrophage2

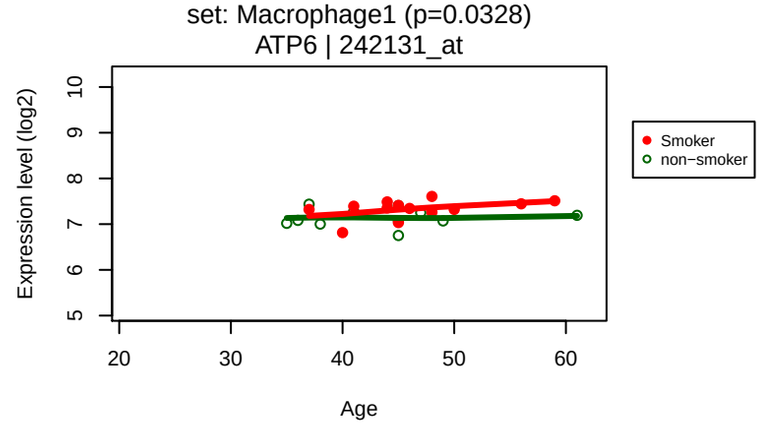

set: SmallAirway1 (p=0.3119)
ATP6 | 242131_at
Expression level (log2)
Age
Smoker
non−smoker

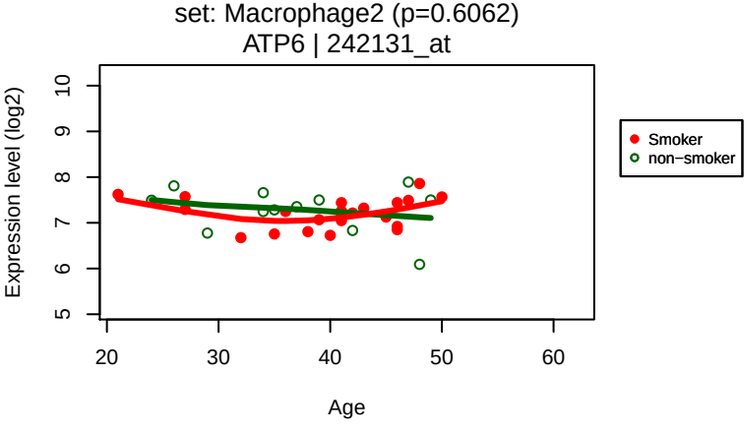

set: SmallAirway2 (p=0.1376)
ATP6 | 242131_at
Expression level (log2)
Age
Smoker
non−smoker

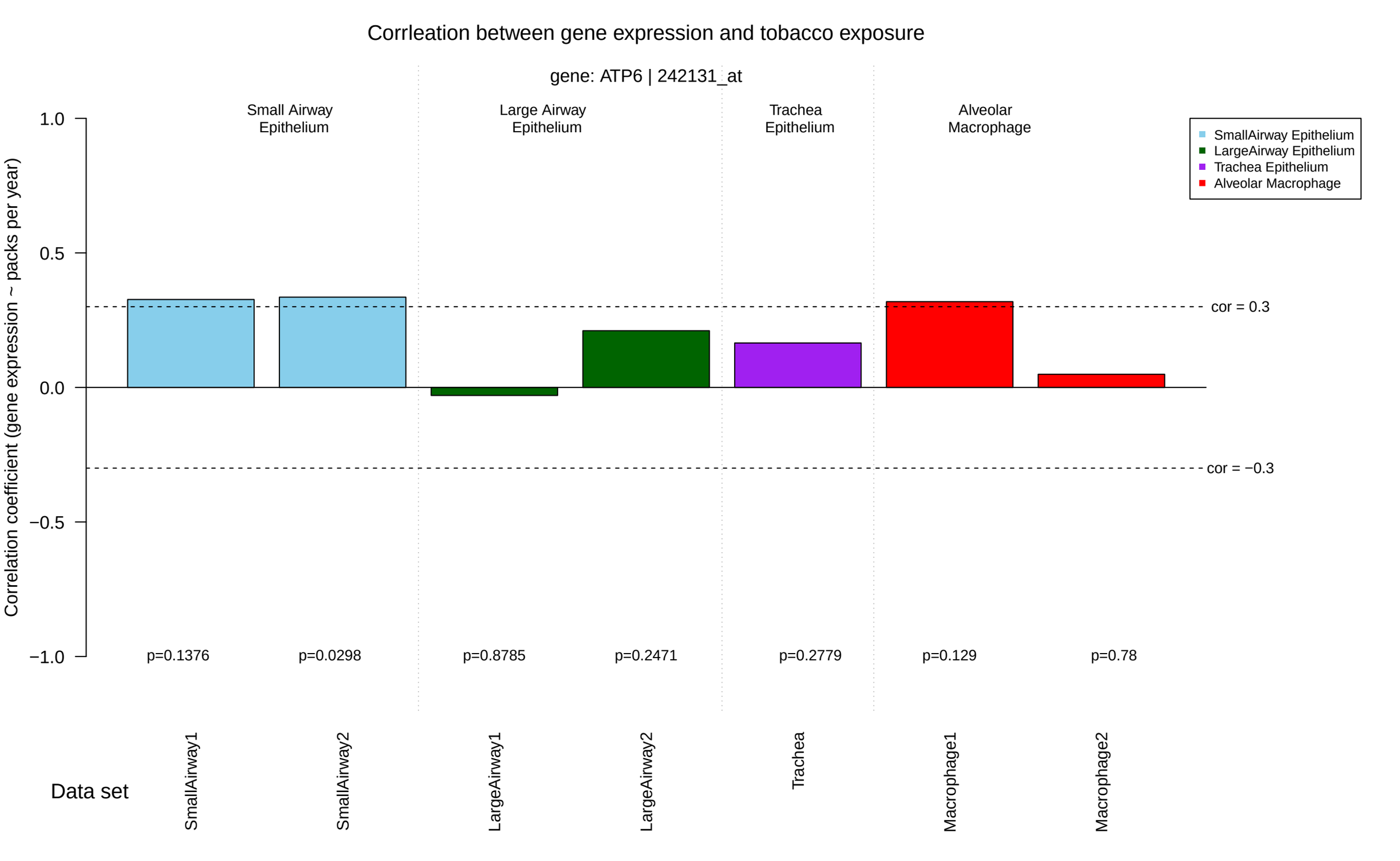

set: LargeAirway1 (p=0.0973)
ATP6 | 242131_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2803)
ATP6 | 242131_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5154)
ATP6 | 242131_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0328)
ATP6 | 242131_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6062)
ATP6 | 242131_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP6 | 242131_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1376
p=0.0298
p=0.8785
p=0.2471
p=0.2779
p=0.129
p=0.78
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2